



In hot summer, we use earthern pots to drink cold water. The pot reduces the temperature of water inside it. Does the earthern pot act as a refrigerator? No. cyclic process is the basic necessity for heat engine or refrigerator. In earthern pot, the cooling process is not due to any cyclic process. The cooling occurs due to evaporation of water molecules which oozes out through pores of the pot. Once the water molecules evaporate, they never come back to the pot. Even though the heat flows from cold water to open atmosphere, it is not a violation of second law of thermodynamics. The water inside the pot is an open thermodynamic system, so the entropy of water + surrounding always increases.

SUMMARY

- Heat is energy in transit which flows from hot object to cold object. However it is not a quantity.
- Work is a process to transfer energy from one object to another object.
- Temperature is a measure of hotness of the object. It determines the direction of the flow of heat.
- The ideal gas law is $PV = NkT$ or $PV = \mu RT$. The Ideal gas law holds for only at thermodynamic equilibrium. For non-equilibrium process, it is not valid.
- Heat capacity is the amount of heat energy required to increase the object's temperature by 1°C or 1K. It is denoted by S.
- Specific heat capacity is the amount of heat energy required to increase the 1 kg of object's temperature by 1°C or 1K. It is denoted by s.
- Molar specific heat capacity is the amount of heat energy requires to increase the 1 mole of substance's temperature by 1°C or 1K. It is denoted by C.
- Thermal expansion is a tendency of an object to change its shape, area, and volume due to change in temperature.
- Water has an anomalous behavior of expansion.
- Latent heat capacity is the amount of heat energy required to change the phase of the substance.
- Calorimetry is the measurement of the amount of heat energy released or absorbed by a thermodynamic system during the heating process.
- Heat transfers in three different modes: conduction, convection and radiation
- Stefan-Boltmann law: $E = \sigma T^4$ and Wien's law: $\lambda_{max} T = b$
- Thermodynamic equilibrium: thermal, mechanical and chemical equilibrium
- Thermodynamic variables : Pressure, temperature, volume, internal energy and entropy

UNIT-8(XI-Physics_Vol-2).indd 153 02-01-2023 15:26:23